

CrabTransformer Training Report

Scenario: REAL | Loss: Tweedie | Lag: 5 Years | Current Spawner: Yes

— Training Configuration —

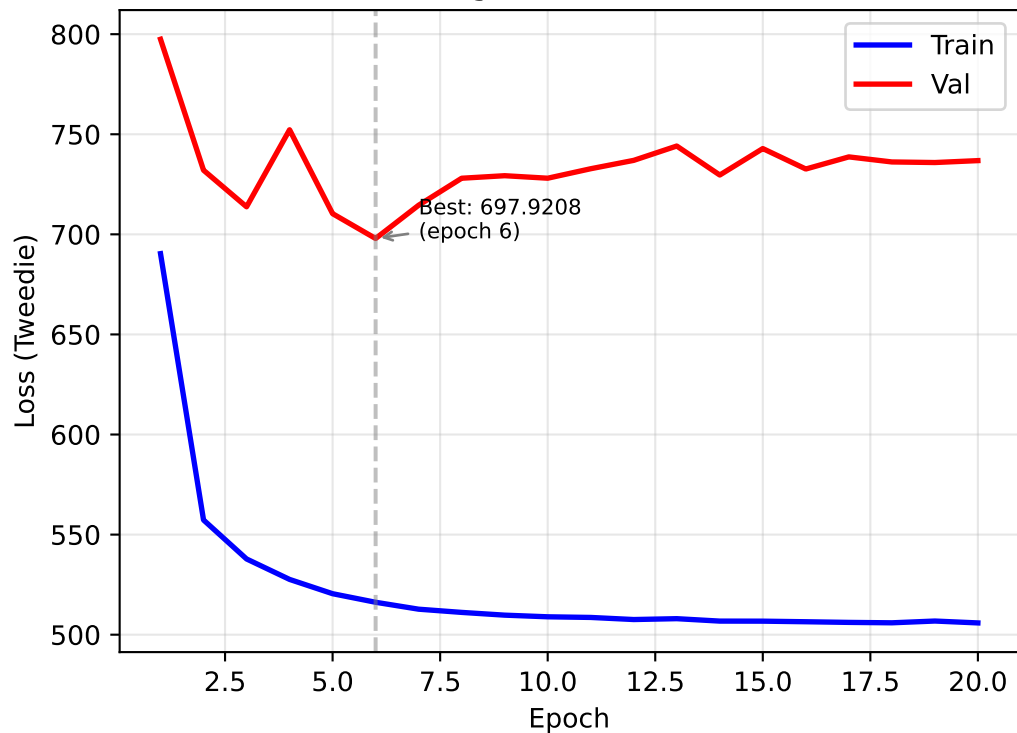
epochs_run: 20 best_val_epoch: 6
best_val_loss: 697.9208 final_train_loss: 505.8813
bias_correction (MSE): 1.0000
include_current_spawner: True

— Model Architecture —

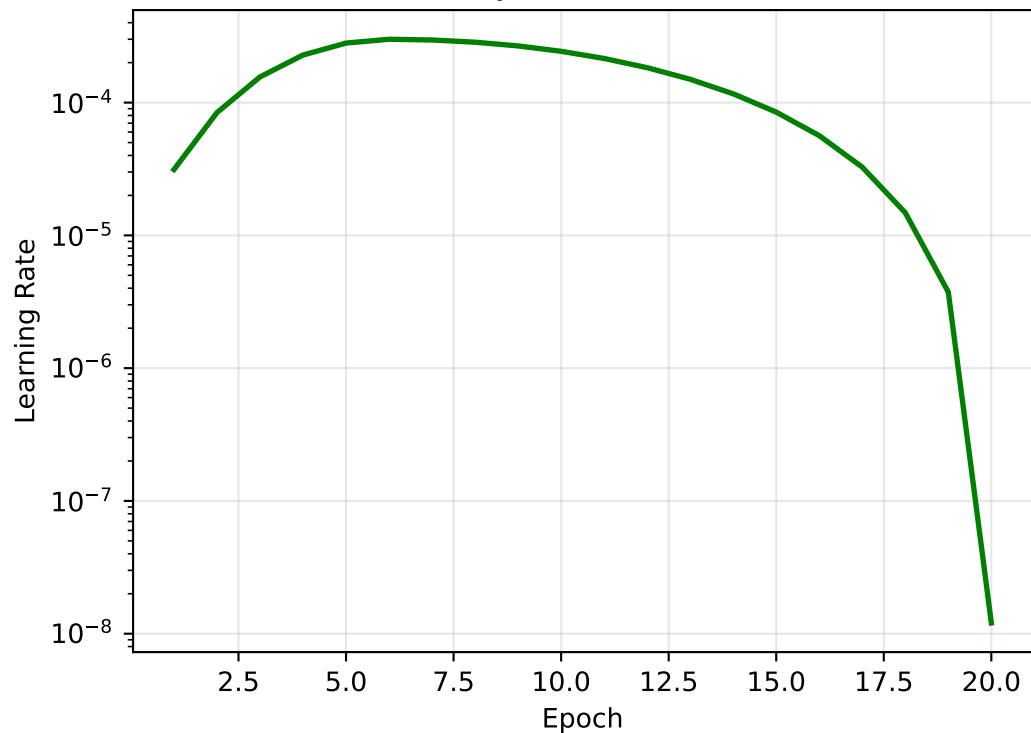
embed_dim: 128 num_heads: 4 num_layers: 3
d_ff: 512 dropout: 0.2 patch_size: 5
train/val/test years: 21/6/4

Training Curves: REAL | Lag 5 | Tweedie | w/ Spawner(t)

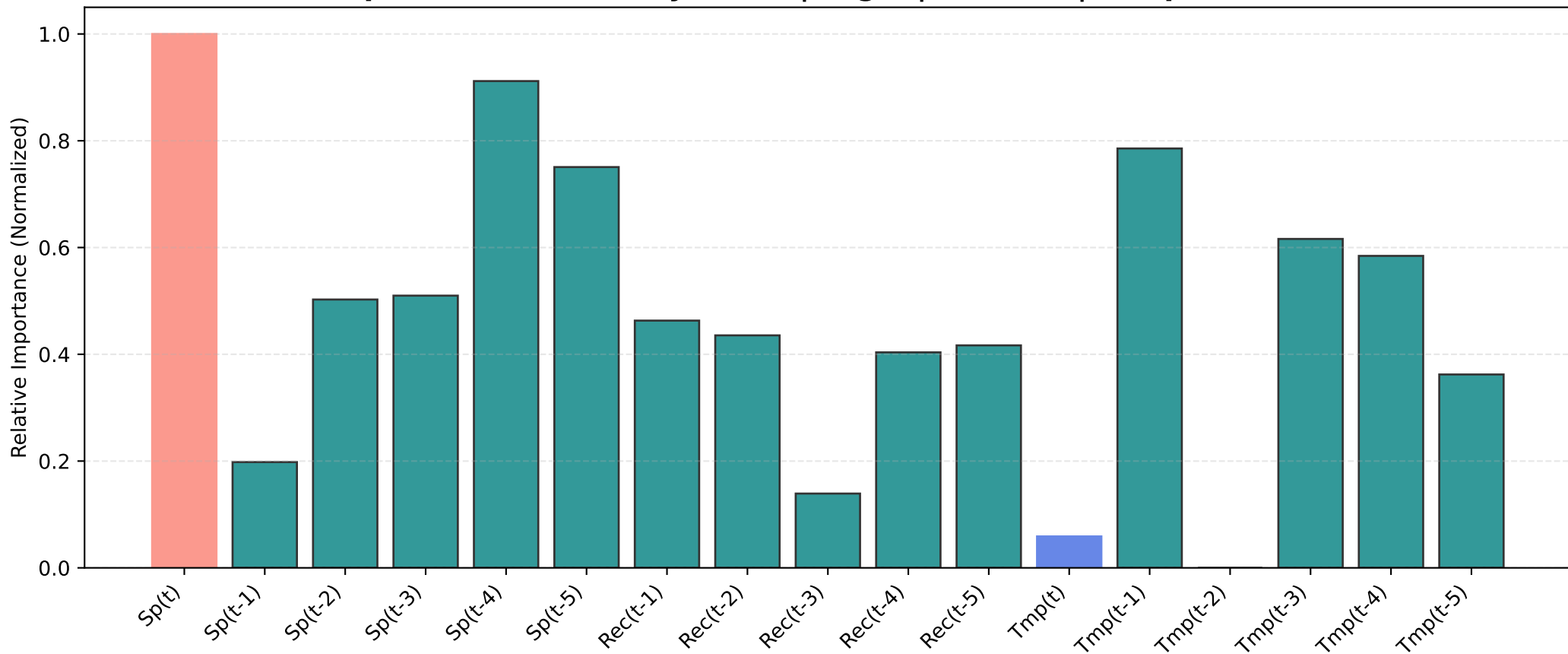
Training & Validation Loss



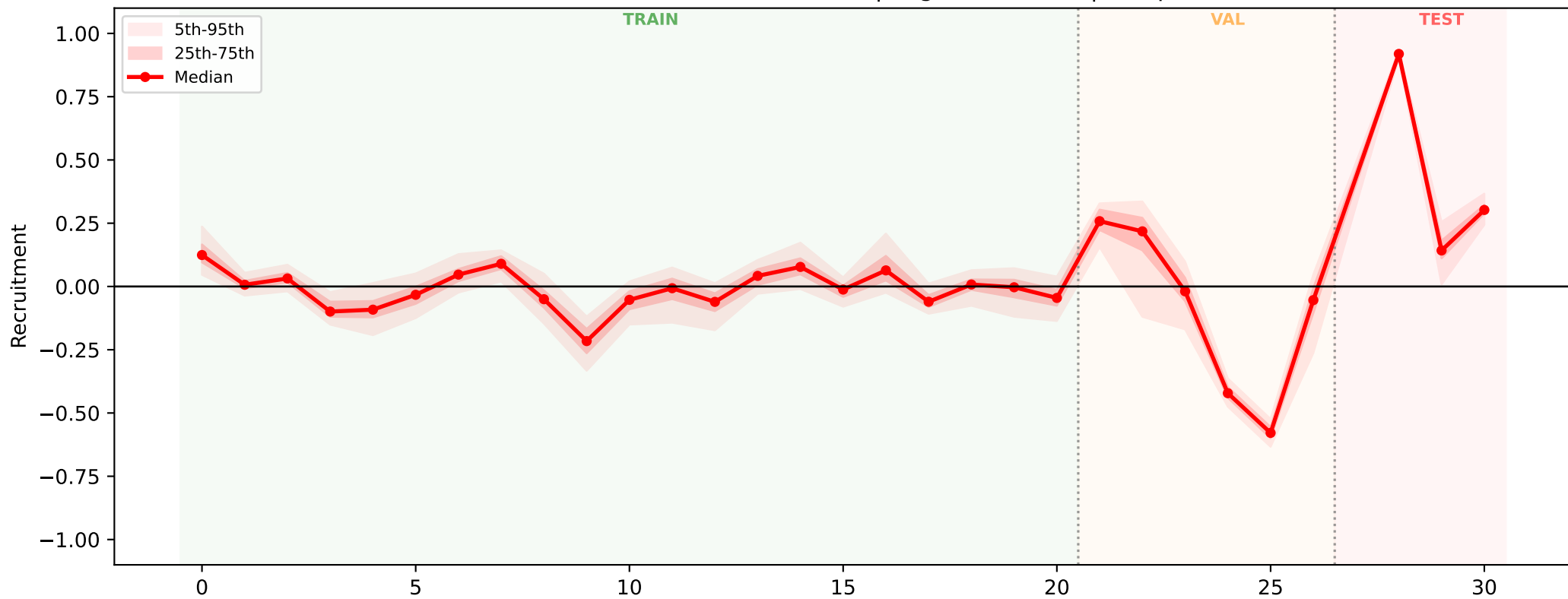
OneCycleLR Schedule



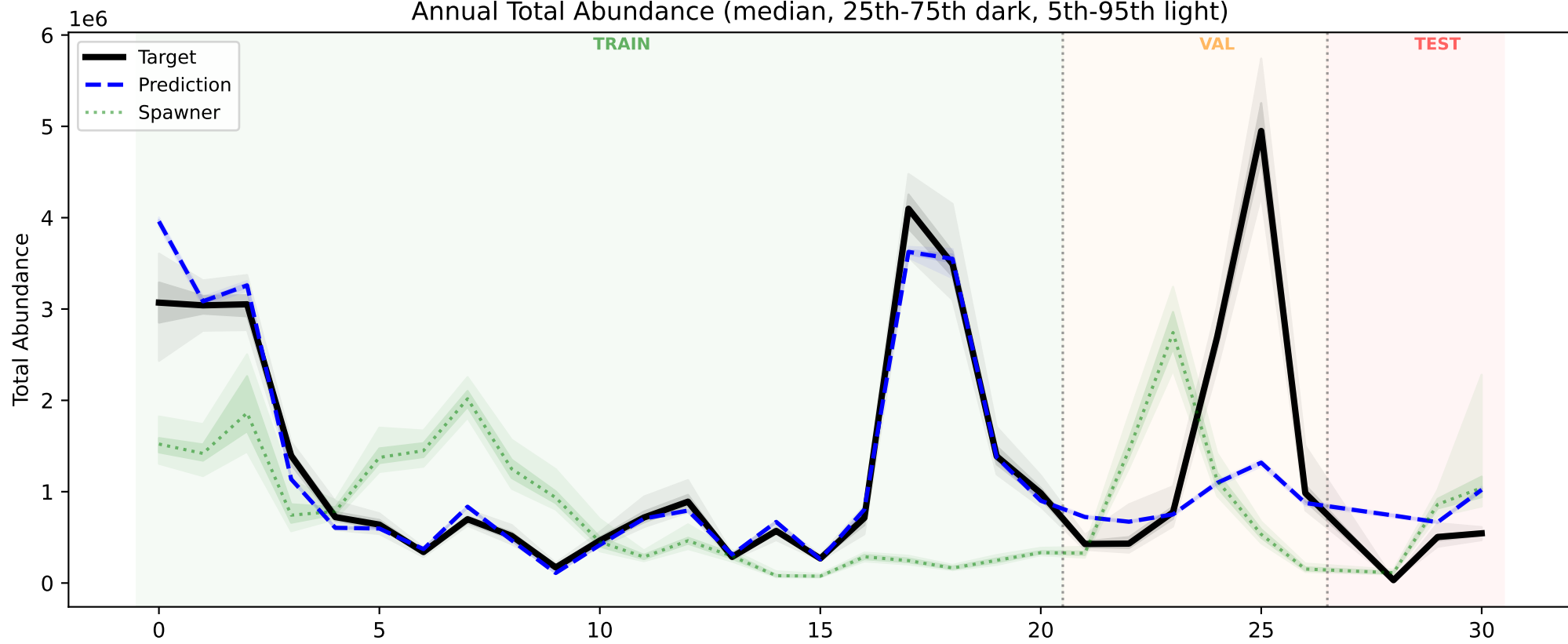
Input Channel Priority: REAL | Lag 5 | Tweedie | w/ Spawner(t)



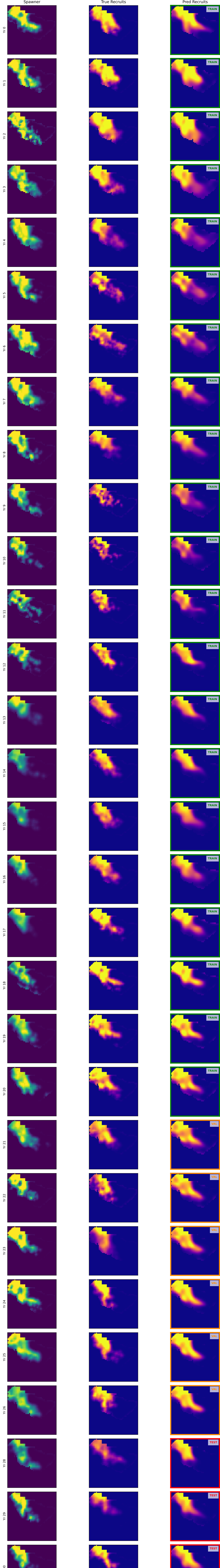
Annual Recruitment Deviations: REAL | Lag 5 (Tweedie) | w/ Spawner(t)



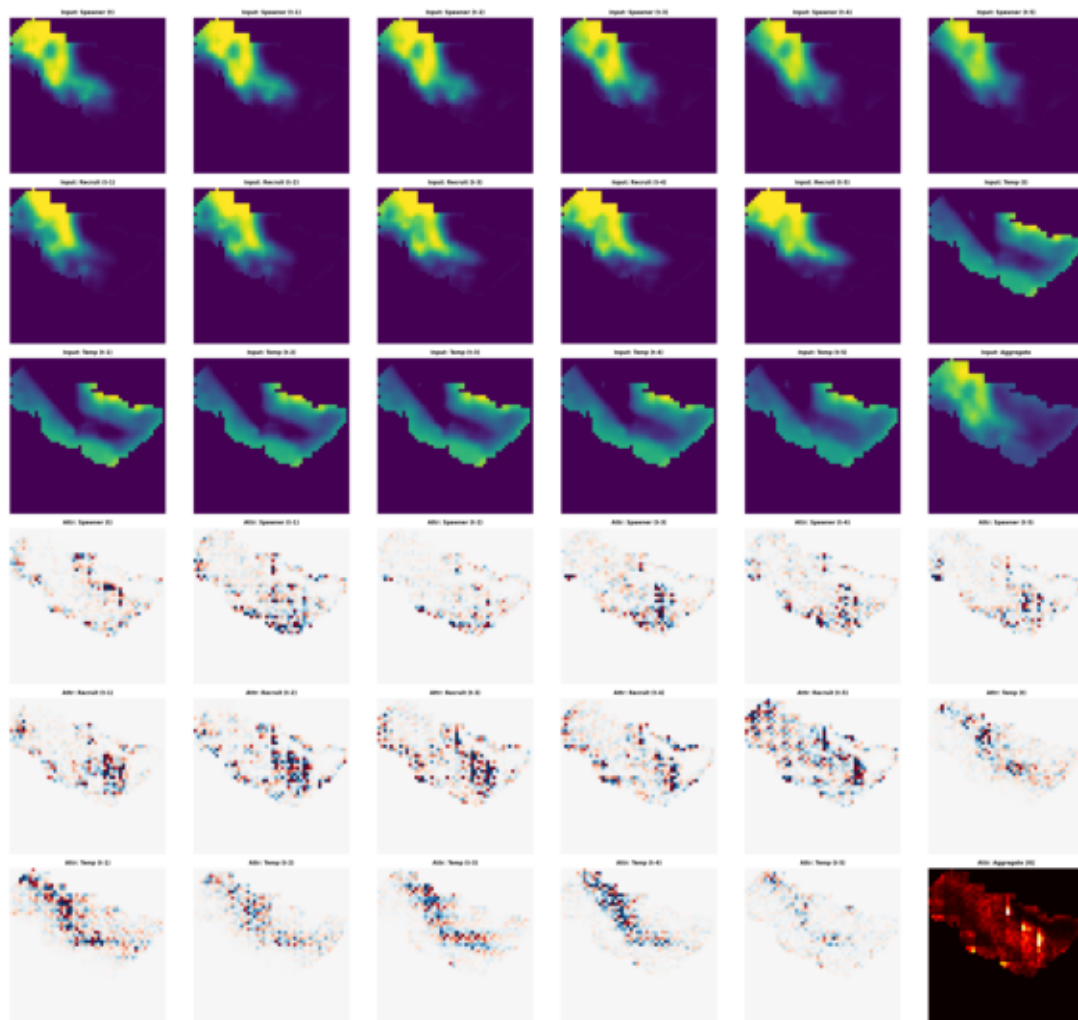
Annual Total Abundance (median, 25th-75th dark, 5th-95th light)



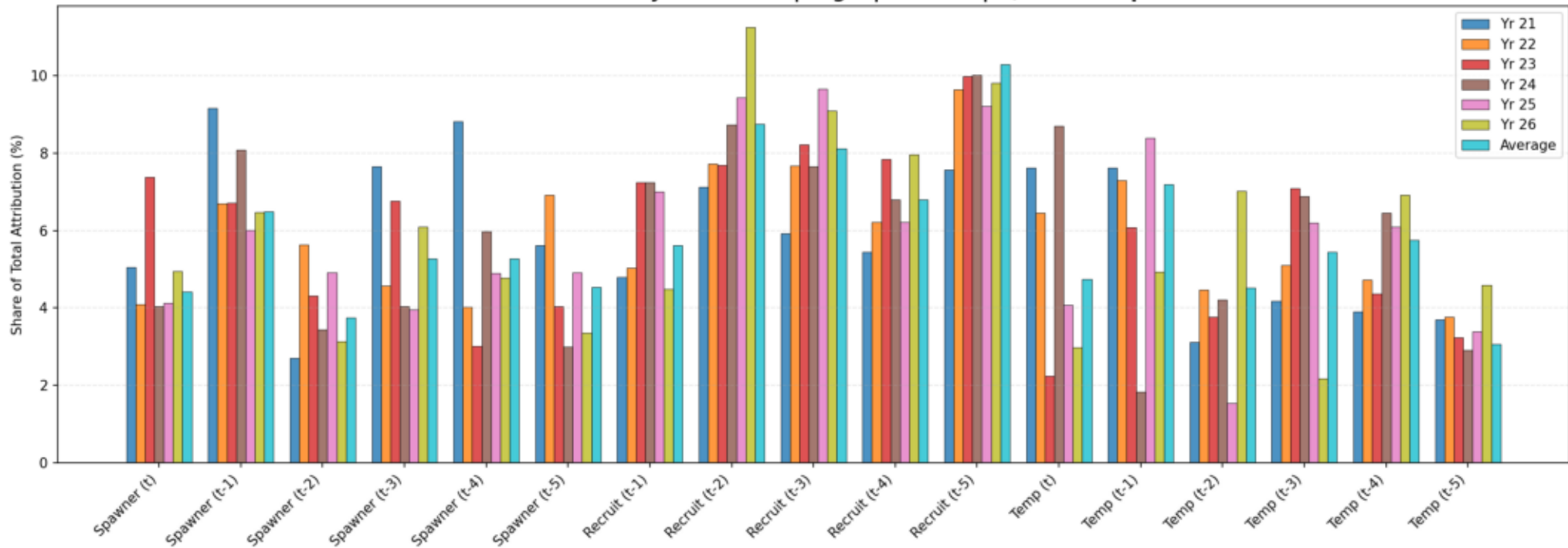
Spatial Trajectory: REAL | Lag 5 | Tweedie | w/ Spawner(t)



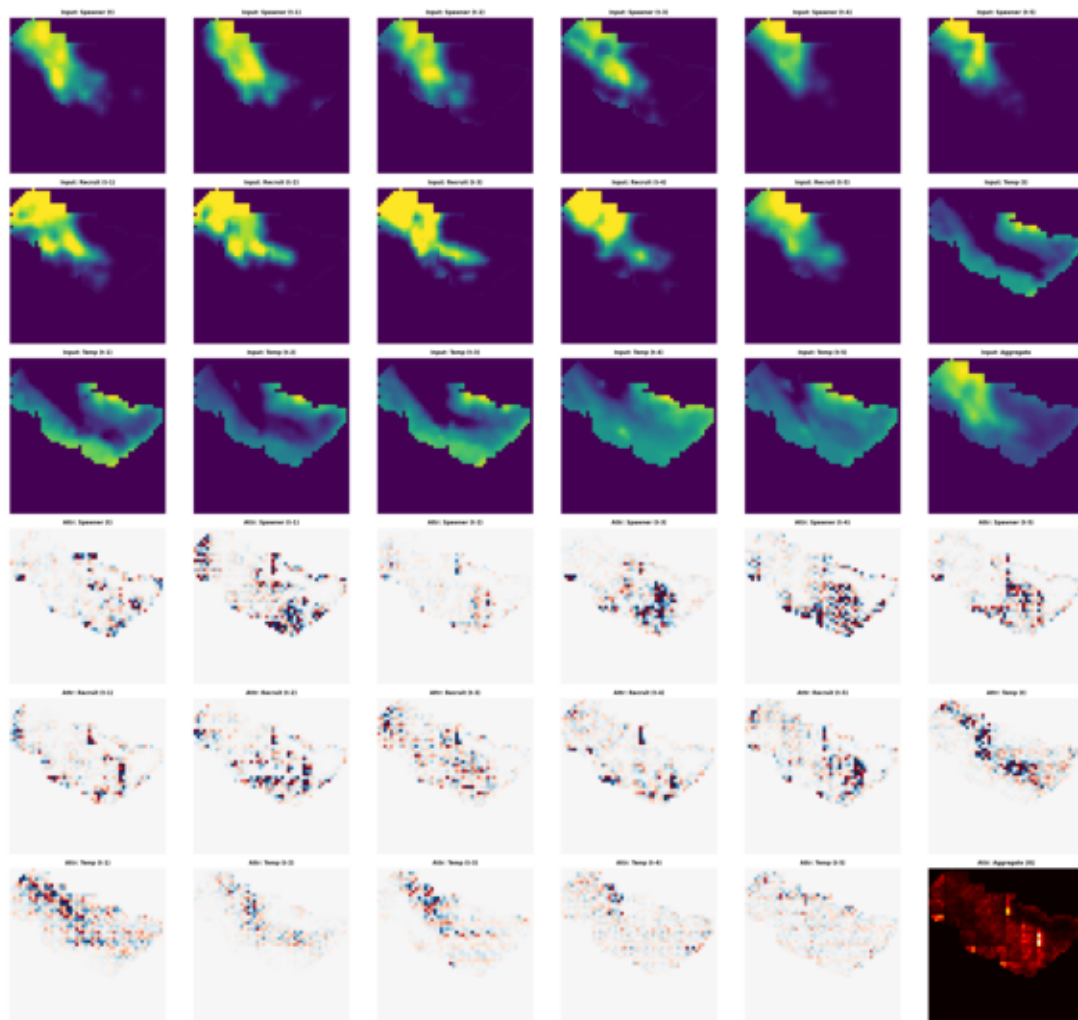
Year AVG — REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



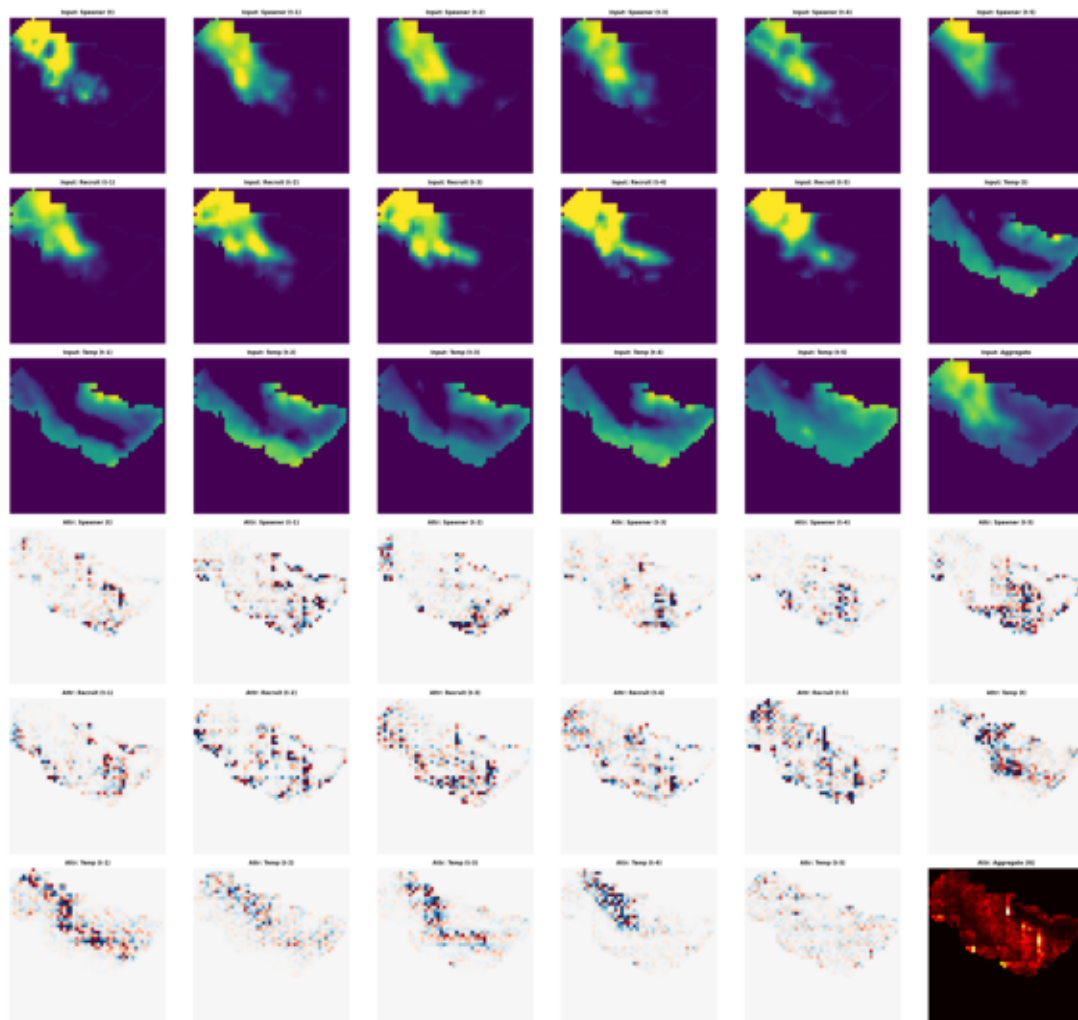
Channel Attribution by Year: REAL | Lag 5 | Tweedie | w/ Current Spawner



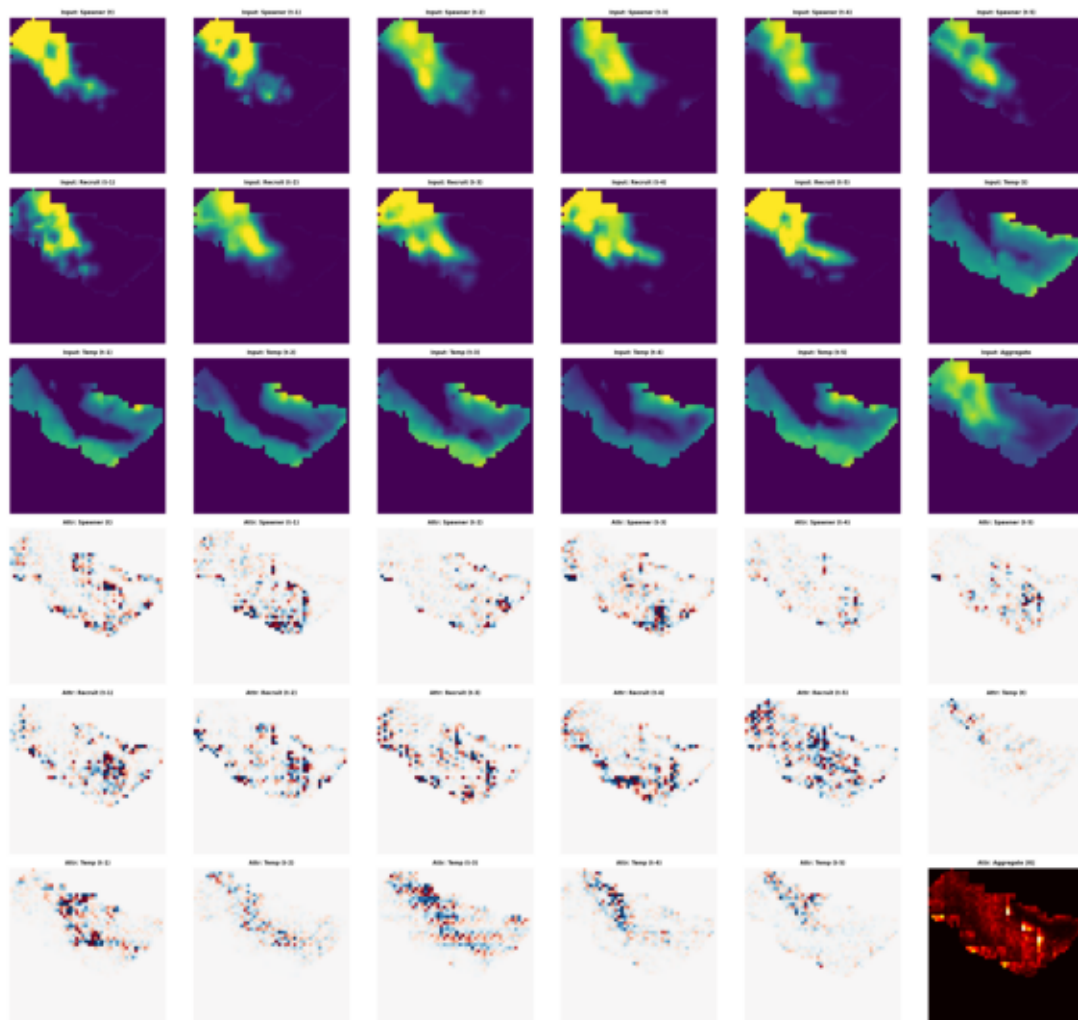
Year 21 — REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



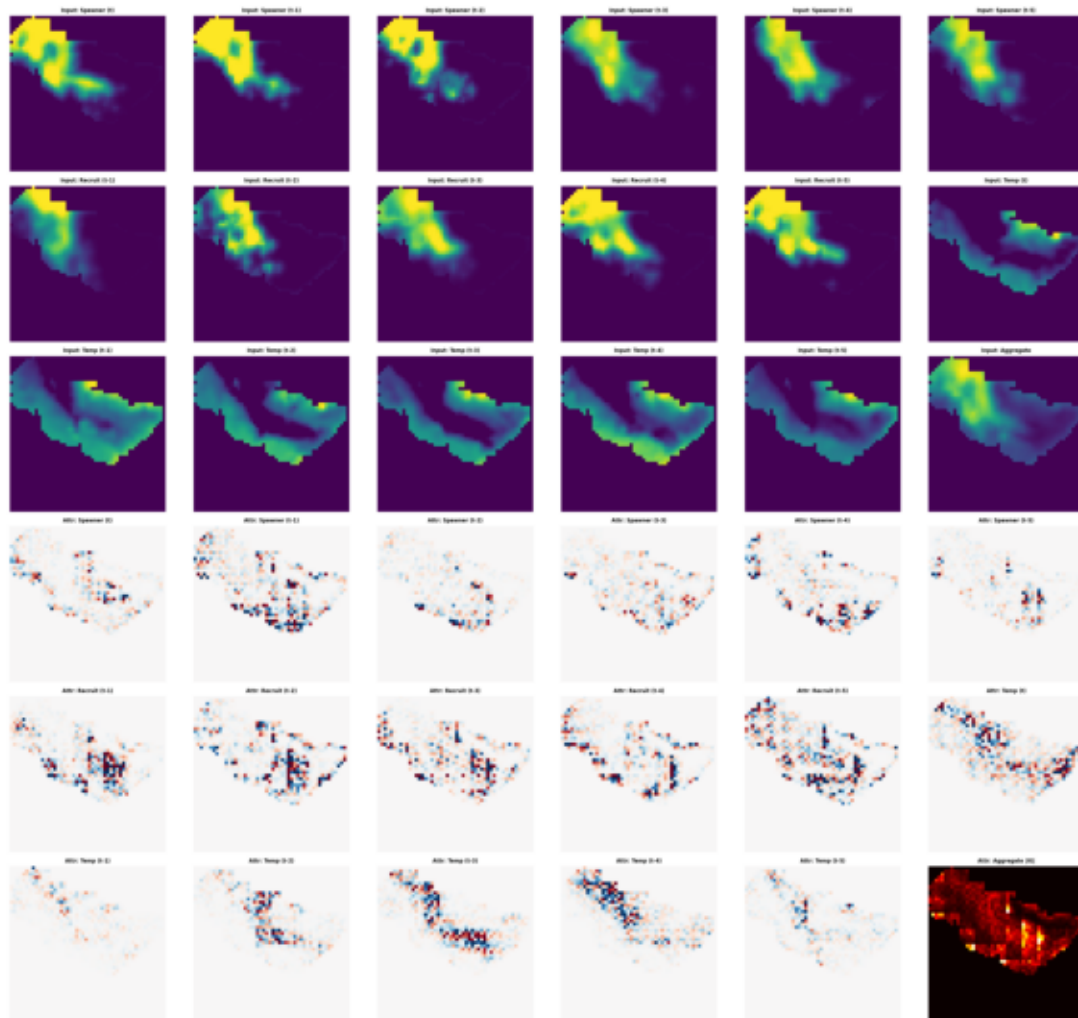
Year 22 -- REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



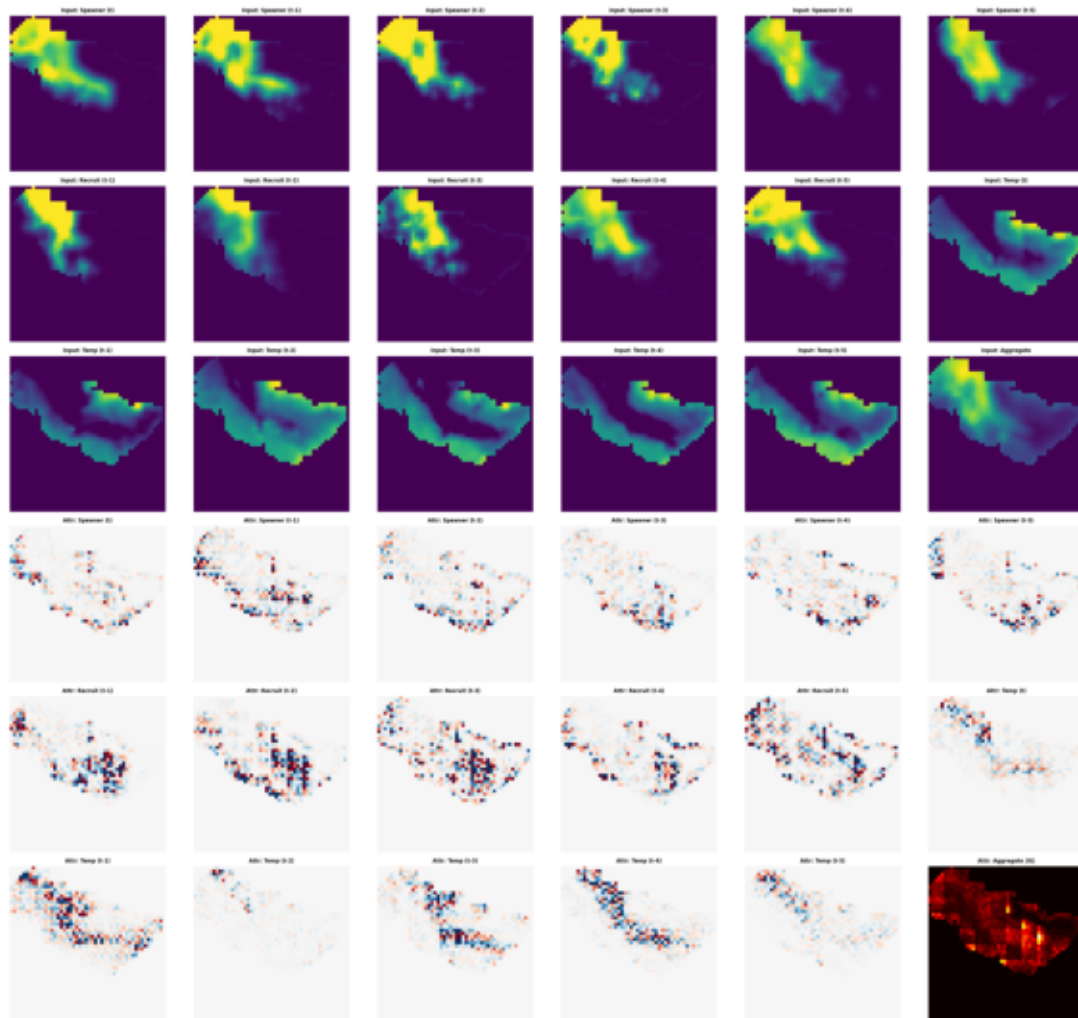
Year 23 -- REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



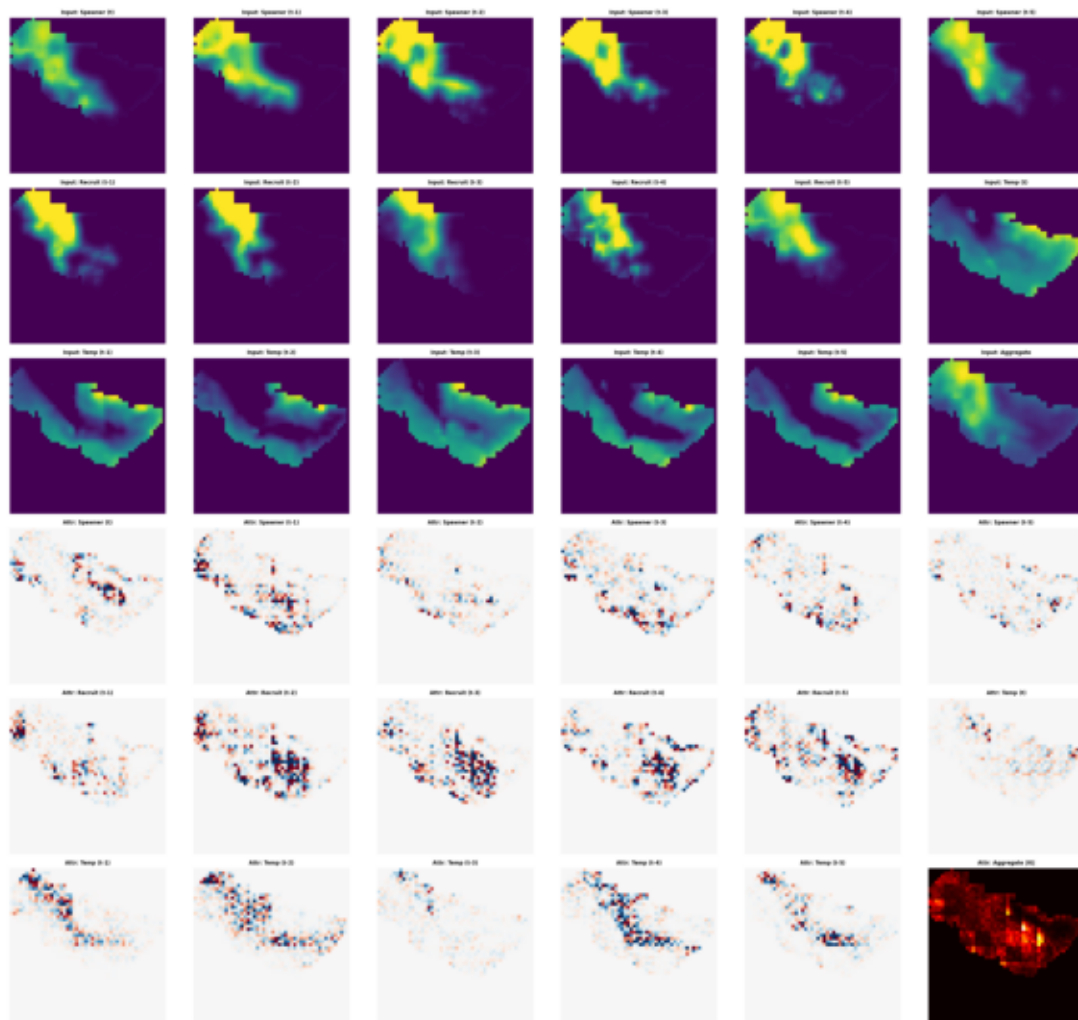
Year 24 -- REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 25 -- REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 26 -- REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Summary Statistics: REAL | Lag 5 | Tweedie | w/ Spawner(t)

Phase	Pred_Mean	Tgt_Mean	Pred_SD	Pred_P50	Pred_P95	Abundance_Capture	MAE	Spearman
TRAIN	1494.4	1480.3	5899.3	0.0	8300.5	1.010	504.9	0.933
VAL	1021.3	1950.8	2476.9	0.0	6284.4	0.524	1758.9	0.804
TEST	911.6	406.3	2979.6	0.0	5522.2	2.244	703.3	0.814

Detailed Metrics (R-style): REAL | Lag 5 | Tweedie | w/ Spawner(t)

Phase	RMSE	RMSE_unbias	Bias	Bias_%	Spearman	Pearson	Pct_Error	Zero_F1	TopK_Acc	TopK_Jaccard	Quant_Err	Skill_MSE	Skill_MAE
TRAIN	1304.7	1291.8	14.2	1.8%	0.940	0.963	-1.8%	0.947	0.883	0.793	0.38	0.900	0.814
VAL	5734.5	5611.7	-929.5	2.9%	0.809	0.717	-7.5%	0.871	0.642	0.479	1.05	0.204	0.413
TEST	2365.0	2297.6	505.3	5.4%	0.821	0.658	824.0%	0.915	0.625	0.463	0.85	-236.642	-4.499